An introduction to deep-learning-based methods for optimization and control of PDEs

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Universidad Politécnica de Cartagena
https://github.com/fperiago/pinn_deeponet_for_beginners

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Outline of the course

- **Part I: Machine Learning basis.** Where and why Machine-Learning-based methods may be useful in the numerical approximation of PDEs-based models?

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- **Part IV: Numerical implementation via DeepXDE.** A Python library for scientific machine learning and physics-informed learning

Part I

Machine Learning Basis

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$$\hat{\mathcal{R}}_n(f) = \frac{1}{n}\sum_{i=1}^{n}\left(f(\boldsymbol{\theta}; \boldsymbol{x}_i) - f^*(\boldsymbol{x}_i)\right)^2, \quad f \in \mathcal{H}_m. \tag{1}$$

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The overall objective is to minimize the **generalization error**

$$\mathcal{R}(f) = \mathbb{E}_{\boldsymbol{x}\sim\mathbb{P}}\left(f(\boldsymbol{\theta};\boldsymbol{x}_i) - f^*(\boldsymbol{x}_i)\right)^2, \quad f\in\mathcal{H}_m, \tag{2}$$

with $\mathbb{P}$ the (unknown) distribution of $\boldsymbol{x}$.

Analyzing the performance of a ML algorithm

What separates ML from optimization is that we want the **generalization error**, also called the **test error**, to be low as well. The factors determining how well a machine learning algorithm will perform are its ability to:

- Make the training error small, and
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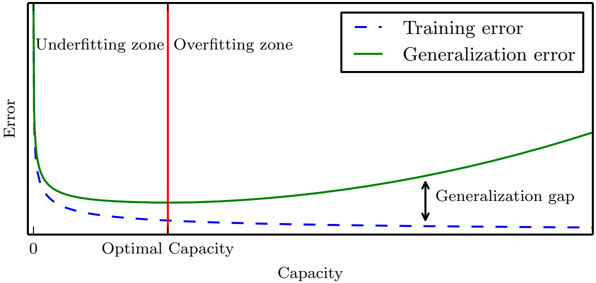
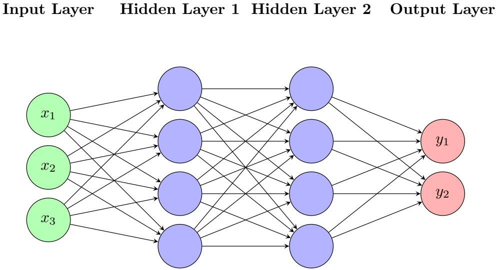


Hypothesis space: an example

A canonical example of an hypothesis space $\mathcal{H}_m$ (or a neural network architecture) is the so-called **Multi-Layer Perceptron (MLP)**.



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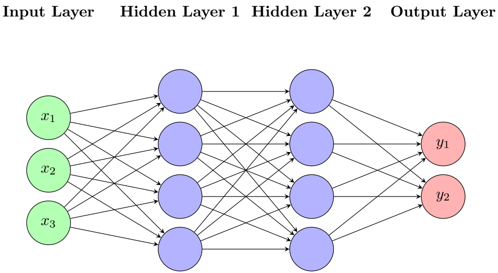

To each input $\boldsymbol{x} \in \mathbb{R}^d$ it associates the output $\boldsymbol{y} = f_m(\boldsymbol{x}) := \boldsymbol{x}^m$ defined by

$$\begin{cases} \text{input layer:} & \boldsymbol{x}^0 = \boldsymbol{x}, \\ \text{hidden layers:} & \boldsymbol{x}^{k+1} = \sigma\left(\omega^{k+1}\boldsymbol{x}^k + b^{k+1}\right) \quad \text{for } k = 0, 1, \cdots, m-2 \\ \text{output layer:} & \boldsymbol{x}^m = \omega^m \boldsymbol{x}^{m-1} + b^m, \end{cases}$$

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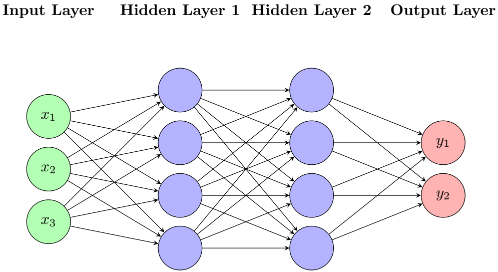

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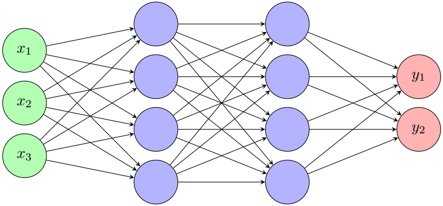

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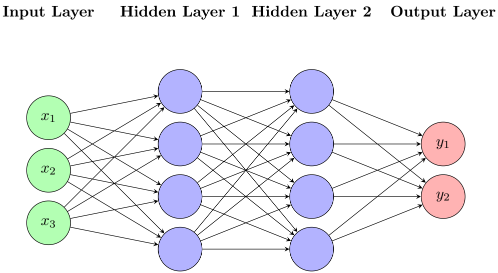

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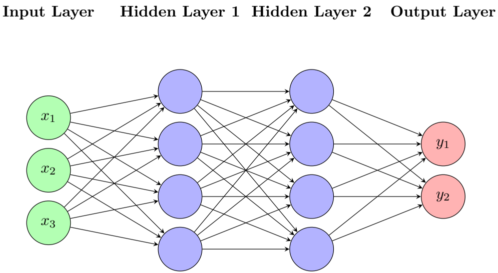

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- σ is a fixed nonlinear **activation function**

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By abuse of notation, $\sigma : \mathbb{R}^d \to \mathbb{R}^d$ is defined component-wise by

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Common choices include *rectifiers* such as ReLU: $\sigma(s) = \max\{s, 0\}$, and *sigmoids* such as $\sigma(s) = \tanh(s)$.

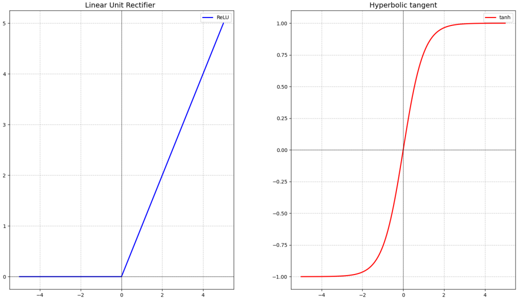

Figure: Linear Unit Rectifier (left) and hyperbolic tangent (right).

Optimization algorithm

Algorithm 1: Adaptive with Moment (Adam)

Requieres: learning rate ϵ (0.001) and decreasing rates for the moments: ρ_1 y ρ_2 ($\rho_1 = 0.9$, $\rho_2 = 0.999$).

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while *stopping condition is not satisfied* **do**

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Algorithm 7: Adaptive with Moment (Adam)

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- Update the optimization variable:
$$\theta_i^{(k+1)} \leftarrow \theta_i^{(k)} - \epsilon \frac{\hat{v}_i^{(k+1)}}{\sqrt{\hat{w}_i^{(k+1)}} + \delta}, \quad i = 1, \dots, r$$
- $k \leftarrow k + 1$

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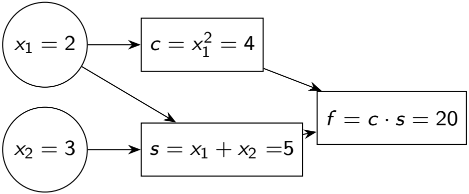

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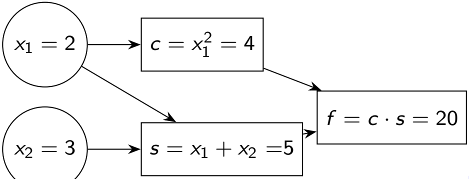
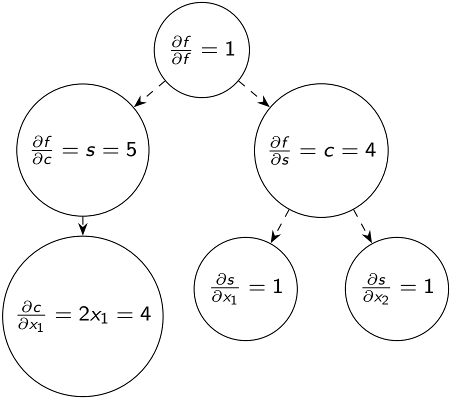


Backpropagation



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- Boltzmann kinetic equations ($d = 6$)

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Typically, $m, n, t \to \infty$ and $d >> 1$.

Examples where d is large include:

- radiactive transport equation ($d \geq 5$)
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- nonlinear Black- Scholes equation for pricing derivatives

More situations that lead to very large d:

- turbulence modeling,
- plasticity models,
- multiscale,
- multiphysics,
- etc.

Deep Learning opens a door to deal with real-world problems

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The heart of the matter for the difficulties described above is our limited ability to handle functions of many variables, and this is exactly where **machine learning** *can make a difference.*

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Machine learning is a promising tool to deal with high-dimensional problems

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Part II

Physics Informed Neural Networks (PINNs)

Main objectives and references

Goals

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- Study the basis of PINNs algorithm for solving a variety of PDEs-based mathematical models

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References

Raisi, M., Perdikaris, P. and Karniadakis, G: Physics-informed neural networks: *A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations,* J. Comput. Physics **378**, 686-707, 2019.

S Wang, S Sankaran, H Wang, P Perdikaris:*An expert's guide to training physics-informed neural networks.* arXiv:2308.08468, 178, 2023.

Lu, L., Meng, X., Mao, Z. and Karniadakis, G.:*DeepXDE: A Deep Learning Library for Solving Differential Equations,* SIAM Review, **63** (1), 208-228, 2021.

García-Cervera, C.J.; Kessler, M.; Periago, F.: *Control of partial differential equations via physics-informed neural networks.* J. Optim. Theory Appl. **196** (2), 391-414, 2023.

A toy model: null control of the wave equation

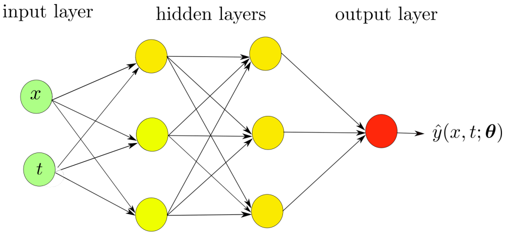
$$\begin{cases} y_{tt} - \Delta y = 0, & \text{in } Q_T \\ y(x,0) = y^0(x), & \text{in } \Omega \\ y_t(x,0) = y^1(x) & \text{in } \Omega \\ y(x,t) = 0, & \text{on } \Gamma_D \times (0,T) \\ y(x,t) = u(x,t) & \text{on } \Gamma_C \times (0,T) \end{cases}$$

Goal: Compute $u(x,t)$ such that

$$y(x,T) = y_t(x,T) = 0 \quad x \in \Omega.$$

Step 1: Neural network

A surrogate $\hat{y}(x,t;\boldsymbol{\theta})$ of the state variable $y(x,t)$ is constructed as



$$\begin{cases} \text{input layer:} & \mathcal{N}^0(\boldsymbol{x}) = \boldsymbol{x} = (x,t) \in \mathbb{R}^{d+1} \\ \text{hidden layers:} & \mathcal{N}^\ell(\boldsymbol{x}) = \sigma\left(\boldsymbol{W}^\ell \mathcal{N}^{\ell-1}(\boldsymbol{x}) + \boldsymbol{b}^\ell\right) \in \mathbb{R}^{N_\ell}, \quad \ell = 1, \cdots, L-1 \\ \text{output layer:} & \hat{y}\left(\boldsymbol{x};\boldsymbol{\theta}\right) = \mathcal{N}^L(\boldsymbol{x}) = \boldsymbol{W}^L \mathcal{N}^{L-1}(\boldsymbol{x}) + \boldsymbol{b}^L \in \mathbb{R} \end{cases}$$

- $\mathcal{N}^\ell : \mathbb{R}^{d_{in}} \to \mathbb{R}^{d_{out}}$ is the ℓ layer with N_ℓ neurons,
- $\boldsymbol{W}^\ell \in \mathbb{R}^{N_\ell \times N_{\ell-1}}$ and $\boldsymbol{b}^\ell \in \mathbb{R}^{N_\ell}$ are, respectively, the weights and biases so that $\boldsymbol{\theta} = \left\{\boldsymbol{W}^\ell, \boldsymbol{b}^\ell\right\}_{1\leq\ell\leq L}$ are the parameters of the neural network, and
- σ is an activation function, e.g. $\sigma(s) = \tanh(s)$

Step 2: Training dataset

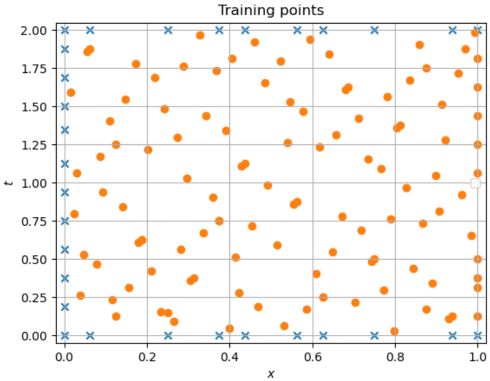


Figure: Illustration of a training dataset (based on Sobol points) in the domain $Q_2 = (0,1) \times (0,2)$. Interior points are marked with circles and boundary points in blue color. $\mathbf{x}_j = (x_j, t_j)$ are the features.

Numerical approximation using PINNs

Step 3: **Loss function.** **Labels equal zero**

$$
\begin{aligned}
\mathcal{L}_{\text{int}}\left(\boldsymbol{\theta}; \mathcal{T}_{\text{int}}\right) &= \sum_{j=1}^{N_{\text{int}}} w_{j,\text{int}} \left|\hat{y}_{tt}(\boldsymbol{x}_j;\boldsymbol{\theta}) - \Delta \hat{y}(\boldsymbol{x}_j;\boldsymbol{\theta})\right|^2, && \boldsymbol{x}_j \in \mathcal{T}_{\text{int}} \\
\mathcal{L}_{\Gamma_D}\left(\boldsymbol{\theta}; \mathcal{T}_{\Gamma_D}\right) &= \sum_{j=1}^{N_b} w_{j,b} \left|\hat{y}(\boldsymbol{x}_j;\boldsymbol{\theta})\right|^2, && \boldsymbol{x}_j \in \mathcal{T}_{\Gamma_D} \\
\mathcal{L}^{\text{pos}}_{t=0}\left(\boldsymbol{\theta}; \mathcal{T}_{t=0}\right) &= \sum_{j=1}^{N_0} w_{j,0} \left|\hat{y}(\boldsymbol{x}_j;\boldsymbol{\theta}) - y^0(\boldsymbol{x}_j)\right|^2, && \boldsymbol{x}_j \in \mathcal{T}_{t=0} \\
\mathcal{L}^{\text{vel}}_{t=0}\left(\boldsymbol{\theta}; \mathcal{T}_{t=0}\right) &= \sum_{j=1}^{N_0} w_{j,0} \left|\hat{y}_t(\boldsymbol{x}_j;\boldsymbol{\theta}) - y^1(\boldsymbol{x}_j)\right|^2, && \boldsymbol{x}_j \in \mathcal{T}_{t=0} \\
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\end{aligned}
$$

where $w_{j,\text{int}}$, $w_{j,b}$, $w_{j,0}$ and $w_{j,T}$ are the weights of suitable quadrature rules.

$$
\begin{aligned}
\mathcal{L}\left(\boldsymbol{\theta}; \mathcal{T}\right) &= \lambda_1 \mathcal{L}_{\text{int}}\left(\boldsymbol{\theta}; \mathcal{T}_{\text{int}}\right) \\
&\quad + \lambda_2 \mathcal{L}_{\Gamma_D}\left(\boldsymbol{\theta}; \mathcal{T}_{\Gamma_D}\right) \\
&\quad + \lambda_3 \mathcal{L}^{\text{pos}}_{t=0}\left(\boldsymbol{\theta}; \mathcal{T}_{t=0}\right) + \lambda_4 \mathcal{L}^{\text{vel}}_{t=0}\left(\boldsymbol{\theta}; \mathcal{T}_{t=0}\right) \\
&\quad + \lambda_5 \mathcal{L}^{\text{pos}}_{t=T}\left(\boldsymbol{\theta}; \mathcal{T}_{t=T}\right) + \lambda_6 \mathcal{L}^{\text{vel}}_{t=T}\left(\boldsymbol{\theta}; \mathcal{T}_{t=T}\right).
\end{aligned}
$$

Numerical approximation using PINNs

Step 4: Training process

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathcal{L}\left(\boldsymbol{\theta}; \mathcal{T}\right).$$

The approximation $\hat{u}(t;\boldsymbol{\theta}^*)$ of the control $u(x,t)$ is

$$\hat{u}(x,t;\boldsymbol{\theta}^*) = \hat{y}(x,t;\boldsymbol{\theta}^*), \quad x \in \Gamma_C,\ 0 \leq t \leq T.$$

Numerical approximation using PINNs

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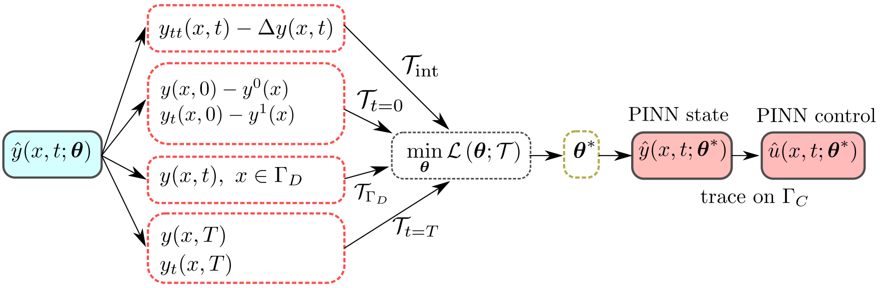
To sump up:



Why MLP is a suitable prediction model ?

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Let us consider the hypothesis space of single-layer neural nets

$$\mathcal{H}_m := \left\{ y_m(\boldsymbol{x}) := \sum_{i=1}^{m} a_i \sigma\left(\boldsymbol{\omega}_i \boldsymbol{x} + b_i\right) : \boldsymbol{x}, \boldsymbol{\omega}_i \in \mathbb{R}^{d+1}, a_i, b_i \in \mathbb{R} \right\}.$$

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Theorem (Pinkus Universal Approximation Theorem)

Let $f \in C^k(\mathbb{R}^{d+1})$. Assume that the activation function $\sigma \in C^k(\mathbb{R})$ is not a polynomial. Then, for any compact set $K \subset \mathbb{R}^{d+1}$ and any $\varepsilon > 0$ there exists $m \in \mathbb{N}$ and $y_m \in \mathcal{H}_m$ such that

$$\max_{\boldsymbol{x} \in K} |D^{\ell} f(\boldsymbol{x}) - D^{\ell} y_m(\boldsymbol{x})| \leq \varepsilon$$

for all multiindex $\ell \leq k$. Moreover, each $a_i = a_i(f)$ is a continuous linear functional defined on K.

Pinkus, A.: *Approximation theory of the MLP model in neural networks* **Acta numer. 8**, 143-195, 1999.

Why is so crucial that the activation function not to be a polynomial?

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If σ is a polynomial of degree m, then $\sigma(\omega x + b)$ is a polynomial of total degree at most m. Thus, $\mathcal{H}_m$ is the space of all polynomials of degree m, which is not dense in $C(K)$, $K \subset \mathbb{R}$ compact.

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Conversely, let $\sigma \in C^{\infty}(\mathbb{R})$. Consider the space

$$\mathcal{N}(\sigma; \mathbb{R}, \mathbb{R}) = \operatorname{span}\{\sigma(wx + b), \quad w, b \in \mathbb{R}\}.$$

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A classical result in Calculus states that if $\sigma\in C^{\infty}$ on any open interval and is not a polynomial thereon, then there exists a point $b^{\star}$ in that interval such that $\sigma^{(k)}(b^{\star})\neq 0$ for all $k=0,1,2,\cdots$.

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$$\frac{\sigma((w + h)x + b^\star) - \sigma(wx + b^\star)}{h} \in \mathcal{N}(\sigma; \mathbb{R}, \mathbb{R})$$

and so

$$\frac{d}{dw}\sigma(wx + b^\star)|_{w=0} = x\sigma'(b^\star) \in \overline{\mathcal{N}(\sigma; \mathbb{R}, \mathbb{R})}.$$

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Since $\sigma^k(b^{\star}) \neq 0$ for all k, then $\overline{\mathcal{N}(\sigma; \mathbb{R}, \mathbb{R})}$ contains all polynomials. By Weierstrass theorem, $\mathcal{N}(\sigma; \mathbb{R}, \mathbb{R})$ is dense in $C(K)$ for any compact set $K \subset \mathbb{R}$.

Estimates on generalization error for the null control of the wave equation

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Training error

$$
\begin{aligned}
\mathcal{E}_{\text{train}} := {} & \mathcal{E}_{\text{train, int}} + \mathcal{E}_{\text{train, boundary}} + \mathcal{E}_{\text{train, initialpos}} + \mathcal{E}_{\text{train, initialvel}} \\
& + \mathcal{E}_{\text{train, finalpos}} + \mathcal{E}_{\text{train, finalvel}},
\end{aligned}
$$

$$
\begin{cases}
\mathcal{E}_{\text{train, int}} & = \left(\mathcal{L}_{\text{int}}\left(\boldsymbol{\theta}^*; \mathcal{T}_{\text{int}}\right)\right)^{1/2} \\
\mathcal{E}_{\text{train, boundary}} & = \left(\mathcal{L}_{\Gamma_D}\left(\boldsymbol{\theta}^*; \mathcal{T}_{\Gamma_D}\right)\right)^{1/2} \\
\mathcal{E}_{\text{train, initialpos}} & = \left(\mathcal{L}_{t=0}^{\text{pos}}\left(\boldsymbol{\theta}^*; \mathcal{T}_{t=0}\right)\right)^{1/2} \\
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\mathcal{E}_{\text{train, finalpos}} & = \left(\mathcal{L}_{t=T}^{\text{pos}}\left(\boldsymbol{\theta}^*; \mathcal{T}_{t=T}\right)\right)^{1/2} \\
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\end{cases}
$$

Estimates on generalization error for the null control of the wave equation

Training error

$$
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\mathcal{E}_{\text{train}} \quad &:= \mathcal{E}_{\text{train, int}} + \mathcal{E}_{\text{train, boundary}} + \mathcal{E}_{\text{train, initialpos}} + \mathcal{E}_{\text{train, initialvel}} \\
&+ \mathcal{E}_{\text{train, finalpos}} + \mathcal{E}_{\text{train, finalvel}},
\end{aligned}
$$

$$
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\end{array}
\right.
$$

Generalization error **for control and state**

$$
\left\{
\begin{array}{l}
\mathcal{E}_{\text{gener}}(u) := \|u - \hat{u}\|_{L^2(\Gamma_C;(0,T))} \\
\mathcal{E}_{\text{gener}}(y) := \|y - \hat{y}\|_{C\left(0,T;L^2(\Omega)\right) \cap C^1\left(0,T;H^{-1}(\Omega)\right)}
\end{array}
\right.
$$

Approximation theory and convergence analysis

Theorem (Estimates on generalization error)

Assume that both $y, \hat{y} \in C^2\left(\overline{Q_T}\right)$. Then

$$\begin{aligned}
\mathcal{E}_{gener}(u) \quad & \leq C\Big(\mathcal{E}_{train,\ int} + C_{q_{int}}^{1/2} N_{int}^{-\alpha_{int}/2} \\
& +\mathcal{E}_{train,\ boundary} + C_{qb}^{1/2} N_b^{-\alpha_b/2} \\
& +\mathcal{E}_{train,\ initialpos} + C_{qip}^{1/2} N_0^{-\alpha_{ip}/2} \\
& +\mathcal{E}_{train,\ initialvel} + C_{qiv}^{1/2} N_0^{-\alpha_{iv}/2} \\
& +\mathcal{E}_{train,\ finalpos} + C_{qfp}^{1/2} N_T^{-\alpha_{fp}/2} \\
& +\mathcal{E}_{train,\ finalvel} + C_{fv}^{1/2} N_T^{-\alpha_{fv}/2}\Big),
\end{aligned}$$

where $C = C(\Omega, T)$, and consequently $C = C(d)$ also depends on the spatial dimension d. A similar estimate holds for the state variable. Moreover, training errors converge to zero as the size of the NN and the number of training points go to infinity.

García-Cervera, C., Kessler, M., Periago, F.: *Control of Partial Differential Equations via Physics-Informed Neural Networks* **J. Optim. Th. Appl. 196**, 391–414, 2023.

Recommendations

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  1. Compute

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  $$\begin{cases} \hat{\lambda}_{ic} = \frac{\|\nabla_\theta \mathcal{L}_{ic}(\theta)\| + \|\nabla_\theta \mathcal{L}_{bc}(\theta)\| + \|\nabla_\theta \mathcal{L}_r(\theta)\|}{\|\nabla_\theta \mathcal{L}_{ic}(\theta)\|} \\ \hat{\lambda}_{bc} = \frac{\|\nabla_\theta \mathcal{L}_{ic}(\theta)\| + \|\nabla_\theta \mathcal{L}_{bc}(\theta)\| + \|\nabla_\theta \mathcal{L}_r(\theta)\|}{\|\nabla_\theta \mathcal{L}_{bc}(\theta)\|} \\ \hat{\lambda}_r = \frac{\|\nabla_\theta \mathcal{L}_{ic}(\theta)\| + \|\nabla_\theta \mathcal{L}_{bc}(\theta)\| + \|\nabla_\theta \mathcal{L}_r(\theta)\|}{\|\nabla_\theta \mathcal{L}_r(\theta)\|} \end{cases}$$
  2. Update the weights $\lambda = (\lambda_{ic}, \lambda_{bc}, \lambda_r)$ by using a moving average
  $$\lambda_{\text{new}} = \alpha\lambda_{\text{old}} + (1-\alpha)\hat{\lambda}_{\text{new}}$$

- **Random weight factorization** may improve the performance of PINNs:

$$w^{(k,\ell)} = s^{(k,\ell)} v^{(k,\ell)},$$

with $s^{(k,\ell)}$ is a trainable scale factor assigned to each individual neuron.

S. Wang, et al..: *Random weight factorization improves the training of continuous neural representations.* **arXiv:2210.01274,** 2022.

Recommendations

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- **Optimization algorithm.** The Adaptive with Moment (Adam) algorithm is the most widely used to minimise the loss function. To speed up convergence near a local minimum, it is convenient to combine Adam (say, first 20000 iterations) with a quasi-Newton method like L-BFGS.

Shortcomings

- PINN has difficulty in approximating functions that have **steep gradients**. A Residual-based Adaptive Refinement algorithm has been proposed which add more residual points in the locations where the PDE residual is large.

  Lu, L., Meng, X., Mao, Z. and Karniadakis, G.: *DeepXDE: A Deep Learning Library for Solving Differential Equations*, **SIAM Review, 63** (1), 208-228, 2021.

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- **MLPs are biased towards learning low frecuencies functions.** This can be mitigated by using a random Fourier feature embedding like

$$\gamma(x) = (\cos(Bx), \sin(Bx)),$$

  where B_{ij} are sampled from Gaussians. This maps input coordinates into high feecuency signals before passing through the network.

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- **PINNs may violate temporal casuality** when solving time-dependent PDEs. We should partition the temporal domain into M equal sequential segments and introduce $\mathcal{L}_r^i(\theta)$ to denote the PDE residual loss within the i-th segment. The PDE residual becomes $\mathcal{L}_r(\theta) = \sum_{i=1}^{M} \lambda_i \mathcal{L}_r^i(\theta)$.

  S Wang, S Sankaran, H Wang, P Perdikaris:*An expert's guide to training physics-informed neural networks.* **ArXiv:2308.08468,** 2023.

Part III

Deep Operator Network (DeepONet)

Main objectives and references

Goals

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- Given two function spaces X and Y, and an operator

$$\mathcal{G} : X \to Y$$

the main goal is to learn (approximate) the operator $\mathcal{G}$ from a dataset.

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References

Lu, L., Jin, P., Pang, G., Zhang, Z., Karniadakis, G.E.:*Learning nonlinear operators via deeponet based on the universal approximation theorem of operators.* **Nature Machine Intelligence 3**(3), 218-229, 2021.

Lanthaler, S., Mishra, S., Karniadakis, G.E.:*Error estimates for DeepONets: a deep learning framework in infinite dimensions.* **Trans. Math. Appl. 6**(1), 1–144, 2022.

García-Cervera, C.J.; Kessler, M.; Pedregal, P.; Periago, F.: *Universal Approximation of Set-Valued Maps and Application to Control.* **Submitted,** 2025.

Problem setup

For the sake of clarity, we focus on the control problem

$$\begin{cases} y_{tt} - y_{xx} = 0, & \text{in } (0,1) \times (0,2) \\ y(x,0) = y^0(x), & \text{on } (0,1) \\ y_t(x,0) = y^1(x) & \text{on } (0,1) \\ y(0,t) = 0, & \text{on } (0,2) \\ y(1,t) = u(t) & \text{on } (0,2) \\ y(x,2) = y_t(x,2) = 0, & \text{on } (0,1). \end{cases}$$

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Goal: **approximate the controllabilty mapping**

$$\begin{array}{rccl} \mathcal{G}: & L^2(0,1)\times H^{-1}(0,1) & \to & L^2(0,2) \\ & (y^0,y^1) & \mapsto & \mathcal{G}(y^0,y^1) := u \end{array}$$

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where u is the **unique** control of minimal L^2-norm.
The operator $\mathcal{G}$ is well-defined (uniqueness of the control), linear and continuous. Continuity is a consequence of the observability inequality

$$\|u\|_{L^2(0,2)} \leq C\left(\|y^0\|_{L^2(0,1)} + \|y^1\|_{H^{-1}(0,1)}\right)$$

Thus, $\mathcal{G}$ is Lipschitz continuous.

Machine Learning setup

- **Dataset**
  We fix a set of **sensor points** $\{x_1, x_2, \cdots, x_m\} \subset [0,1]$. The information of each selected continuous initial datum (y^0, y^1) is encoded in the vector

$$(y^0(x_1), y^0(x_2), \cdots, y^0(x_m); y^1(x_1), y^1(x_2), \cdots, y^1(x_m)) \equiv y^{\text{initial}}$$

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- **Hypothesis space: the neural network**

  We will use the so-called **DeepONet**, which takes the form

$$\mathcal{N}(\boldsymbol{\theta}; (y^{\text{initial}}(x_j); t)) := \sum_{k=1}^{p} \sum_{i=1}^{n} c_i^k \sigma \left(\sum_{j=1}^{m} \xi_{ij}^k y^{\text{initial}}(x_j) + \theta_i^k \right) \cdot \sigma(w_k \cdot t + \eta_k)$$

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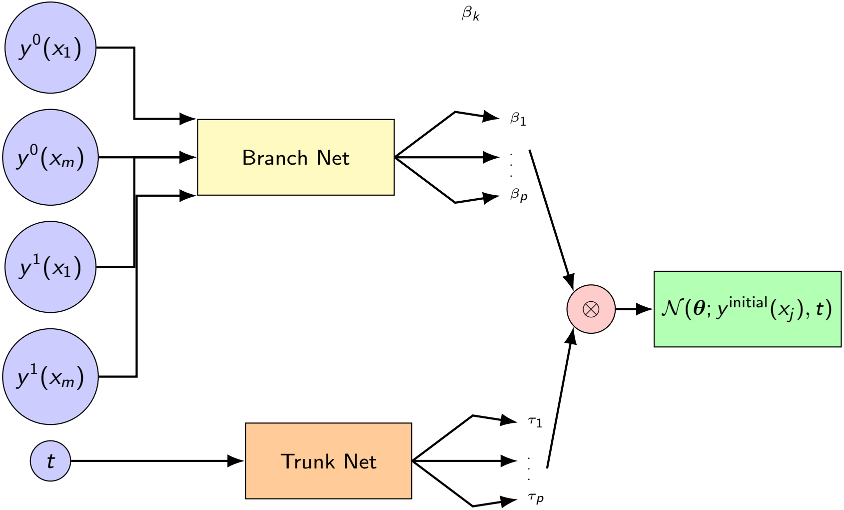
  where $\boldsymbol{\theta} = (c_i^k, \xi_{ij}^k, \theta_i^k, w_k, \eta_k)$ is the set of parameters of the net.
- **Loss function: Mean Squared Error (MSE)**
  $$\text{MSE}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{\ell=1}^{N} |\mathcal{N}(\boldsymbol{\theta}; (y_\ell^{\text{initial}}; t_\ell)) - u_\ell|^2$$

DeepONet's architecture

$$\mathcal{N}(\boldsymbol{\theta}; (y^{\text{initial}}(x_j); t)) := \sum_{k=1}^{p} \underbrace{\sum_{i=1}^{n} c_i^k \sigma \left(\sum_{j=1}^{m} \xi_{ij}^k y^{\text{initial}}(x_j) + \theta_i^k \right)}_{\beta_k} \cdot \underbrace{\sigma(w_k \cdot t + \eta_k)}_{\tau_k}$$



From a different perspective, the approximation scheme may be decomposed as

$$\begin{array}{ccc} X & \xrightarrow{\mathcal{G}} & Y \\ \downarrow \mathcal{E} & & \uparrow \mathcal{R} \\ \mathbb{R}^{2m} & \xrightarrow{\mathcal{A}} & \mathbb{R}^{p} \end{array}$$

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where:

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- $\mathcal{R} : \beta_k(y^{initial}) \mapsto \sum_{k=1}^{p} \beta_k(y^{\text{initial}})\tau_k(t)$ is a *reconstruction operator.*

Thus,

$$\mathcal{G} \approx \mathcal{R} \circ \mathcal{A} \circ \mathcal{E}.$$

Where does this architecture come from ?

Theorem (Universal Approximation Theorem for Functions)

Suppose that $K \subset \mathbb{R}^d$ *is compact,* $U \subset C(K)$ *is compact, and* $\sigma \in C^\infty(\mathbb{R})$ *is not a polynomial. Then, for any* $\varepsilon > 0$ *there exist a positive integer* n*, real numbers* θ_i*,* $\omega_i \in \mathbb{R}^n$*, independent of* $f \in U$*, and constants* $c_i = c_i(f)$ *depending on* f*, such that*

$$|f(x) - \sum_{i=1}^{n} c_i \sigma(\omega_i \cdot x + \theta_i)| < \varepsilon$$

holds for all $x \in K$ *and* $f \in U$*. Moreover, each* $c_i(f)$ *is a continuous linear functional defined on* U*.*

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holds for all $x \in K$ and $f \in U$. Moreover, each $c_i(f)$ is a continuous linear functional defined on U.

Theorem (Universal Approximation Theorem for Functionals)

Suppose that σ is not a polynomial, X is a Banach space, $K \subset X$ is a compact set, V is a compact set in $C(K)$, and $f : V \to \mathbb{R}$ is a continuous functional. Then for any $\varepsilon > 0$, there are a positive integer n, m sensor points $x_1, x_2, \cdots, x_m \in K$, and real constants c_i, θ_i, ξ_{ij}, $1 \leq i \leq n$, $1 \leq j \leq m$, such that

$$|f(y) - \sum_{i=1}^{n} c_i \sigma\left(\sum_{j=1}^{m} \xi_{ij} y(x_j) + \theta_i\right)| < \varepsilon, \quad \text{for all } y \in V.$$

Where does this architecture come from ?

if $\mathcal{G}$ is an operator between two function spaces, from the Universal Approximation Theorem for functions we get

$$\mathcal{G}(y)(t) \approx \sum_{k=1}^{n} c_k\left(\mathcal{G}(y)\right)\sigma(\omega_k \cdot t + \theta_k),$$

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Suppose that σ is not a polynomial, X is a Banach space, $K_1 \subset X$, $K_2 \subset \mathbb{R}^d$ are compact sets, V is a compact set in $C(K_1)$, and $\mathcal{G} : V \to C(K_2)$ is a continuous operator. Then for any $\varepsilon > 0$, there are a positive integers n, p, m sensor points $x_1, x_2, \cdots, x_m \in K_1$, and real constants $c_i^k, \theta_i^k, \xi_{ij}^k, \eta_k$ such that

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Chen, T., Chen, H.: *Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems.* **IEEE Transactions on Neural Networks 6**(4), 911-917, 1995.

DeepONet setup: Dataset

Definition (Data for DeepONet approximation)

Assume that $X \hookrightarrow L^2(D)$, and $Y \hookrightarrow L^2(U)$, for some Banach spaces X, Y. The pair $(\mu, \mathcal{G})$ is said to be **data for DeepONet approximation** provided $\mu \in \mathcal{P}_2(X)$ is Borel measurable with finite second moments, there exists a Borel set $A \subset X$, composed of continuous functions, $\mu(A) = 1$, and $\mathcal{G} : X \to Y$ is a Borel measurable mapping with $\|\mathcal{G}\|_{L^2(\mu)} < \infty$.

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We choose $\mu \in \mathcal{P}\left(C(0,1)\right)$ a probability measure whose probability law is given by a Karhunen-Loève expansion

$$\mu \sim \sum_{j=1}^{\infty} \sqrt{\lambda_j}\xi_j(\omega)\varphi_j(x) \tag{3}$$

We take $\xi_j \sim \mathcal{N}(0,1)$ i.i.d. standard Gaussian variables, and (λ_j, φ_j) are the eigenvalues and normalized eigenfuncions of the operator

$$\mathcal{C}(\phi)(x) = \int_0^1 C(x,x')\phi(x')\,dx', \quad C(x,x') = \sigma^2 \exp\left(-\frac{|x-x'|^2}{2\ell^2}\right)$$

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Thus, we truncate (3) and sample the Gaussian random variables. Remember that by Mercer's theorem $\{\varphi_j\}$ is an ortonormal basis of $L^2(0,1)$.

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The encoding operator $\mathcal{E} : C(0,1) \times C(0,1) \to \mathbb{R}^m \times \mathbb{R}^m$ maps $(y^0, y^1) \mapsto y^{\text{initial}}$

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The encoding operator $\mathcal{E} : C(0, 1) \times C(0, 1) \to \mathbb{R}^m \times \mathbb{R}^m$ maps $(y^0, y^1) \mapsto y^{\text{initial}}$
We repeat this process for a number of initial conditions (y_j^0, y_j^1) and store the corresponding y_{jk}^{initial} in the rows of the so-called matrix of **features.**

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Putting all together, the **training dataset** is

$$\{(y_{jk}^{\text{initial}}; u_j(t)), 1 \leq j \leq N, 1 \leq k \leq 2m\},$$

evaluated at a finite selection of times t. Precisely,

$$\left[\begin{array}{cccc}
y_{1,1}^{\text{initial}} & \cdots & y_{1,2m}^{\text{initial}} & u_1(t_1) \\
\cdots & \cdots & \cdots & \cdots \\
y_{1,1}^{\text{initial}} & \cdots & y_{1,2m}^{\text{initial}} & u_1(t_\ell) \\
\hline
\cdots & \cdots & \cdots & \cdots \\
\cdots & \cdots & \cdots & \cdots \\
\hline
y_{N,1}^{\text{initial}} & \cdots & y_{N,2m}^{\text{initial}} & u_N(t_1) \\
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\end{array}\right]$$

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- **Approximation error:** This is the distance between the Hypothesis space and the operator $\mathcal{G}$ to be approximated, i.e., if $\mathcal{F}$ is a fixed space of DeepONets, then

$$\mathcal{N}_{\mathcal{F}} = \arg\min_{\mathcal{N}\in\mathcal{F}} \mathcal{L}(\mathcal{N}) := \int_{L^2(D)} \int_U |\mathcal{G}(y)(t) - \mathcal{N}_{\mathcal{F}}(y)(t)|^2 \, dt d\mu(y),$$

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- **Generalization (or estimation) error:** We approximate $\mathcal{N}_{\mathcal{F}}$ by using a specific (training) dataset $\mathcal{T}$, and a **empirical loss**. So, we get

$$\mathcal{N}_{\mathcal{T}} = \arg\min_{\mathcal{N}\in\mathcal{F}} \mathcal{L}_M(\mathcal{N}) := \frac{|U|}{M} \sum_{j=1}^{M} |\mathcal{G}(y_j)(t_j) - \mathcal{N}(y_j)(t_j)|^2$$

Thus, $\mathcal{E}_{\text{gener}} := \left(\mathcal{L}(\mathcal{N}_{\mathcal{T}}) - \mathcal{L}(\mathcal{N}_{\mathcal{F}})\right)^{1/2}$.

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- **Optimization error:** The empirical loss is highly nonlinear, non-convex so that we compute a local minimum $\mathcal{N}_M$ of $\mathcal{L}_M$. Optimization error is then

$$\mathcal{E}_{\text{optim}} := \|\mathcal{N}_M - \mathcal{N}_{\mathcal{T}}\|^{1/2}.$$

$$\mathcal{E}_{\text{total}} = \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{gener}} + \mathcal{E}_{\text{optim}}$$

Definition (Curse of dimensionality)

For a given $\varepsilon > 0$, let $\mathcal{N}_\varepsilon$ be a DeepONet such that $\mathcal{E}_{\text{approx}} < \varepsilon$, and

$$\text{size}\left(\mathcal{N}_\varepsilon\right) \sim \mathcal{O}\left(\varepsilon^{-\vartheta_\varepsilon}\right) \quad \text{for some } \vartheta_\varepsilon \geq 0.$$

Our DeepONet approximation, with underlying measure μ, is said to *incurr a curse of dimensionality* if $\lim_{\varepsilon\to 0} \vartheta_\varepsilon = +\infty$ and *breaks the curse of dimensionality* if $\lim_{\varepsilon\to 0} \vartheta_\varepsilon = \overline{\vartheta} < +\infty$.

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Error analysis and the curse of dimensionality

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Yarotsky proved that the approximation of a general Lipschitz function to accuracy ε requires a ReLU network of size[1] $\varepsilon^{-m(\varepsilon)/2}$, with $m(\varepsilon) \to \infty$ as $\varepsilon \to 0$, and hence suffers from the curse of dimensionality.

Yarotsky, D.: *Optimal approximation of continuous functions by very deep relu networks.* **Conference on Learning Theory. PMLR,** 639-649, 2018.

In our setting, m is the number of sensors for the enconding operator $u \mapsto \mathcal{E}(u) = (u(x_1), \cdots, u(x_m))$.

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- bounded linear operators $\mathcal{G} : L^2(D) \to L^2(U)$

Application to our controllability setting

Set $\mathbb{T} = [0, \pi]$. Consider the operator

$$\mathcal{G} : L^2(\mathbb{T}) \times H^{-1}(\mathbb{T}) \to L^2(0, 4\pi), \quad (y^0, y^1) \mapsto \mathcal{G}(y^0, y^1) = u$$

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We take $X = C(\mathbb{T}) \times C(\mathbb{T})$. For the sake of simplicity, we consider the measure $\mu \times \mu$, where μ is the one dimensional measure associated with squared exponential covariance funcion

$$C(x, x') = e^{-\frac{|x-x'|^2}{2\ell^2}}$$

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The key ingredient here is that in the decomposition

$$\begin{array}{ccc} X & \xrightarrow{\mathcal{G}} & Y \\ \downarrow \mathcal{E} & & \uparrow \mathcal{R} \\ \mathbb{R}^{2m} & \xrightarrow{\mathcal{A}} & \mathbb{R}^p \end{array}$$

the error associated with the linear operator $\mathcal{A}$ vanishes since if σ is an ReLU activation function, then

$$\mathcal{A}x + b = \sigma(\mathcal{A}x + b) - \sigma(-(\mathcal{A}x + b)).$$

Application to our controllability setting

Theorem (DeepONet approximation of controllability systems)

Consider the control system for $x \in \Omega \subset \mathbb{R}^d$ *and* $t > 0$*:*

$$\begin{cases} y' + Ay = 1_\omega u \\ y(0) = y^0 \end{cases}$$

and assume that for some $T > 0$ *there existe a unique control* $u \in L^2(\omega \times (0,T))$ *such that* $y(T) = 0$*.*
Consider the operator $\mathcal{G} : y^0 \to u$*. Then, for any fixed* $\varepsilon > 0$ *and* $\sigma > 0$*, there exists a DeepONet* $\mathcal{N} = (\beta, \tau)$ *achieving an approximation error*

$$\mathcal{E}_{approx} = \|\mathcal{G} - \mathcal{N}\|_{L^2(\mu)} \lesssim \varepsilon$$

and such that

$$size(\beta) \lesssim \log(\varepsilon^{-1}), \quad depth(\beta) \lesssim 1, \quad p \sim \log(\varepsilon^{-1})^d, \quad m \sim \log(\varepsilon^{-1})^{2d+\sigma}.$$

García-Cervera, C.J.; Kessler, M.; Pedregal, P.; Periago, F.: *Universal Approximation of Set-Valued Maps and Application to Control.* **Submitted,** 2025.

Estimates for generalization error for DeepONet

Boundedness assumption: there exists $\psi : L^2(D) \to [0, +\infty[$ such that

$$|\mathcal{G}(y)(t)| \le \psi(y), \qquad \sup_{\theta \in [-B,B]^{d_\theta}} |\mathcal{N}_\theta(y)(t)| \le \psi(y), \quad \forall y \in L^2(D), \forall t \in U,$$

and there exists $C, \kappa > 0$ such that

$$\psi(y) \le C\left(1 + \|y\|_{L^2(D)}\right)^\kappa.$$

Lipschitz continuity assumption: there exists $\Phi : L^2(D) \to [0, +\infty[$ such that

$$|\mathcal{N}_\theta(y)(t) - \mathcal{N}_{\theta'}(y)(t)| \le \Phi(y)\|\theta - \theta'\|_{\ell^\infty}, \quad \forall y \in L^2(D), \forall t \in U,$$

and there exists $C, \kappa > 0$ such that

$$\Phi(y) \le C\left(1 + \|y\|_{L^2(D)}\right)^\kappa.$$

Theorem (Bound for generalization error)

Let $\mathcal{N}_{\mathcal{F}}$ and $\mathcal{N}_{\mathcal{T}}$ be as before. If the above two assumptions hold, then

$$\mathbb{E}\left[\mathcal{L}\left(\mathcal{N}_{\mathcal{T}}\right) - \mathcal{L}\left(\mathcal{N}_{\mathcal{F}}\right)\right] \le \frac{C}{\sqrt{N}}\left(1 + Cd_\theta \log\left(CB\sqrt{N}\right)\right)^{2\kappa + \frac{1}{2}} \tag{4}$$

where the constant $C = C(\mu, \psi, \Phi)$ is independent of B and d_θ.

Part IV

Numerical Implementation via DeepXDE

PINN. Experiment 1: the Laplace-Poisson equation

$$\begin{cases} -\Delta u(x) = -2d, & x \in D := (0,1)^d \\ u(x) = \sum_{k=1}^{d} x_k^2, & x \in \partial D \end{cases}$$

This problem has the exact solution

$$u_{\text{exact}}(x) = \sum_{k=1}^{d} x_k^2$$

PINN. Experiment 1: the Laplace-Poisson equation

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Table: L^2- relative error $\frac{\| u_{\text{exact}} - u_{\text{PINN}} \|_{L^2(D)}}{\| u_{\text{exact}} \|_{L^2(D)}}$ for different values of the spatial dimension d and number of training points $N = N_{\text{interior}} + N_{\text{boundary}}$

	$N = 100 + 40$	$N = 1000 + 400$	$N = 10000 + 4000$
$d = 2$	9×10^{-4}	1.2×10^{-4}	7.5×10^{-5}
$d = 3$	1.2×10^{-3}	1.4×10^{-4}	1.3×10^{-3}
$d = 5$	2.5×10^{-2}	7.5×10^{-4}	1.9×10^{-4}
$d = 10$	2.2×10^{-1}	4.2×10^{-3}	2.2×10^{-3}
$d = 20$	3.1×10^{-1}	2.6×10^{-2}	2.5×10^{-3}

PINN. Experiment 1: the Laplace-Poisson equation

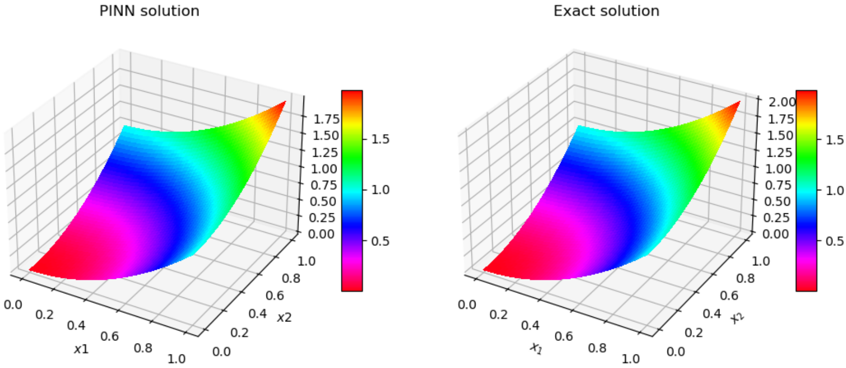

Figure: Experiment 1. Comparison between PINN (or predicted) solution u_{PINN} **(left)**, and exact solution u_{exact} **(right)**. Neural network composed of 3 hidden layers and 50 neurons in each layer. No regularization. Number of training points $N_{\mathrm{interior}} = 100$, $N_{\mathrm{boundary}} = 40$.

PINN. Experiment 2: exact control of the wave equation

Compute $u(t)$ such that the solution $y(x,t)$ of the problem

$$\begin{cases} y_{tt} - y_{xx} = 0, & \text{in } (0,1)\times(0,2) \\ y(x,0) = \sin(\pi x), & \text{in } (0,1) \\ y_t(x,0) = 0 & \text{in } (0,1) \\ y(0,t) = 0, & \text{on } (0,2) \\ y(1,t) = u(t) & \text{on } (0,2) \end{cases}$$

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This problem has an exact solution which can be obtained by means of D'Alembert formula. Indeed, by considering the function

$$\tilde{y^0}(x) = \begin{cases} \sin(\pi x) & -1 \leq x \leq 1 \\ 0 & \text{elsewhere,} \end{cases}$$

the explicit exact state is given by

$$y(x,t) = \frac{1}{2}\left(\tilde{y^0}(x-t) + \tilde{y^0}(x+t)\right), \quad 0 \leq x \leq 1,\ 0 \leq t \leq 2,$$

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and the exact control is

$$u(t) = \begin{cases} \frac{1}{2} y^0(1-t) & 0 \le t \le 1 \\ \\ -\frac{1}{2} y^0(t-1) & 1 \le t \le 2. \end{cases}$$

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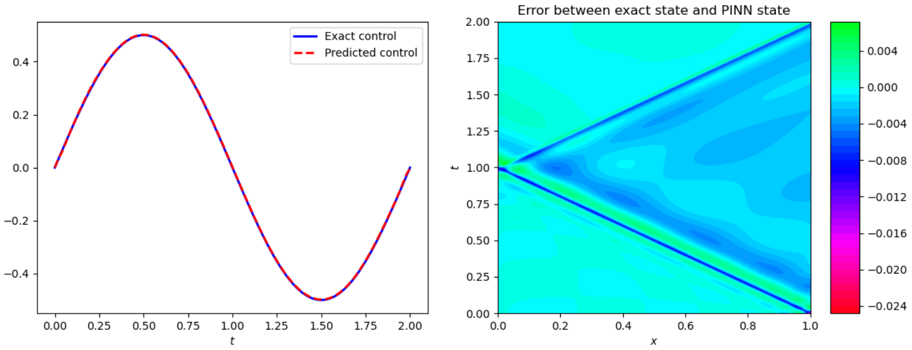

Figure: Experiment 1 (linear wave equation). Comparison between exact control $u(t)$ and PINN (or predicted) control $\hat{u}(t;\boldsymbol{\theta}^*)$ **(left)**, and error between exact state and PINN state, i.e. $y(x,t)-\hat{y}(x,t;\boldsymbol{\theta}^*)$ **(right)**. Neural network composed of 4 hidden layers and 50 neurons in each layer. No regularization. Number of training points $N = 10300$.

DeepONet. Experiment 3: generation of training and test datasets

$$\begin{array}{rll} \mathcal{G}: & L^2(0,1) \times H^{-1}(0,1) & \to L^2(0,2) \\ & (y^0, y^1) & \mapsto \mathcal{G}(y^0, y^1) := u \end{array}$$

where u is the unique control of minimal L^2-norm of the system

$$\begin{cases} y_{tt} - y_{xx} = 0, & \text{in } (0,1) \times (0,2) \\ y(x,0) = y^0(x), & \text{on } (0,1) \\ y_t(x,0) = y^1(x) & \text{on } (0,1) \\ y(0,t) = 0, & \text{on } (0,2) \\ y(1,t) = u(t) & \text{on } (0,2) \\ y(x,2) = y_t(x,2) = 0, & \text{on } (0,1). \end{cases}$$

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where u is the unique control of minimal L^2-norm of the system

$$\begin{cases} y_{tt}-y_{xx}=0, & \text{in } (0,1)\times(0,2) \\ y(x,0)=y^0(x), & \text{on } (0,1) \\ y_t(x,0)=y^1(x) & \text{on } (0,1) \\ y(0,t)=0, & \text{on } (0,2) \\ y(1,t)=u(t) & \text{on } (0,2) \\ y(x,2)=y_t(x,2)=0, & \text{on } (0,1). \end{cases}$$

The **matrix of features** corresponds to initial conditions $(y^0(x),y^1(x))$ evaluated at a number of sensor points $x_j\in(0,1)$. It is computed by sampling a Gaussian random field

$$a(x,\omega)=\sum_{i=1}^{\infty}\sqrt{\lambda_i}e_i(x)\xi_i(\omega), \tag{5}$$

where ξ_i are iid standard Gaussian variables, and $\{\lambda_i,e_i(x)\}_{i=1}^{\infty}$ are the eigenvalues and normalized eigenfuncions of the operator

$$\mathcal{C}(\phi)(x)=\int_0^1 C(x,x')\phi(x')\,dx', \quad C(x,x')=\sigma^2\exp\left(-\frac{|x-x'|^2}{2\ell^2}\right)$$

DeepONet. Experiment 3: generation of training and test datasets

The **vector of labels** corresponds to the exact control, which is given by

$$u(t) = \begin{cases} \frac{1}{2} y^0\,(1-t) + \frac{1}{2}\int_{1-t}^{1} y^1(s)\,ds & 0 \leq t \leq 1 \\ \\ -\frac{1}{2} y^0\,(t-1) + \frac{1}{2}\int_{t-1}^{1} y^1(s)\,ds & 1 \leq t \leq 2. \end{cases} \tag{6}$$

A data point is a triplet $((y^0, y^1), t, \mathcal{G}(y^0, y^1)(t))$.

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$$u(t) = \begin{cases} \frac{1}{2} y^0 (1-t) + \frac{1}{2} \int_{1-t}^{1} y^1(s)\, ds & 0 \leq t \leq 1 \\ \\ -\frac{1}{2} y^0 (t-1) + \frac{1}{2} \int_{t-1}^{1} y^1(s)\, ds & 1 \leq t \leq 2. \end{cases} \tag{6}$$

A data point is a triplet $((y^0, y^1), t, \mathcal{G}(y^0, y^1)(t))$.Moreover, a specific input (y^0, y^1) appears in multiple data points with different values of t. Thus, The **vector of labels** corresponds to the exact control, which is given by

$$u(t) = \begin{cases} \frac{1}{2} y^0 (1-t) + \frac{1}{2} \int_{1-t}^{1} y^1(s)\, ds & 0 \le t \le 1 \\ \\ -\frac{1}{2} y^0 (t-1) + \frac{1}{2} \int_{t-1}^{1} y^1(s)\, ds & 1 \le t \le 2. \end{cases} \tag{6}$$

A data point is a triplet $((y^0, y^1), t, \mathcal{G}(y^0, y^1)(t))$.Moreover, a specific input (y^0, y^1) appears in multiple data points with different values of t. Thus,

$$X_{\text{train}} = \left[\begin{array}{cccc} y_{1,1}^{\text{initial}} & \cdots & y_{1,2m}^{\text{initial}} & t_1 \\ \cdots & \cdots & \cdots & \cdots \\ y_{1,1}^{\text{initial}} & \cdots & y_{1,2m}^{\text{initial}} & t_\ell \\ \hline \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \hline y_{N,1}^{\text{initial}} & \cdots & y_{N,2m}^{\text{initial}} & t_1 \\ \cdots & \cdots & \cdots & \cdots \\ y_{N,1}^{\text{initial}} & \cdots & y_{N,2m}^{\text{initial}} & t_\ell \end{array}\right], \qquad y_{\text{train}} = \left[\begin{array}{c} u_1(t_1) \\ \cdots \\ u_1(t_\ell) \\ \hline \cdots \\ \cdots \\ \hline u_N(t_1) \\ \cdots \\ u_N(t_\ell) \end{array}\right]$$

DeepONet. Experiment 4: fitting the deeponet model

The implementation in DeepXDE is as follows:

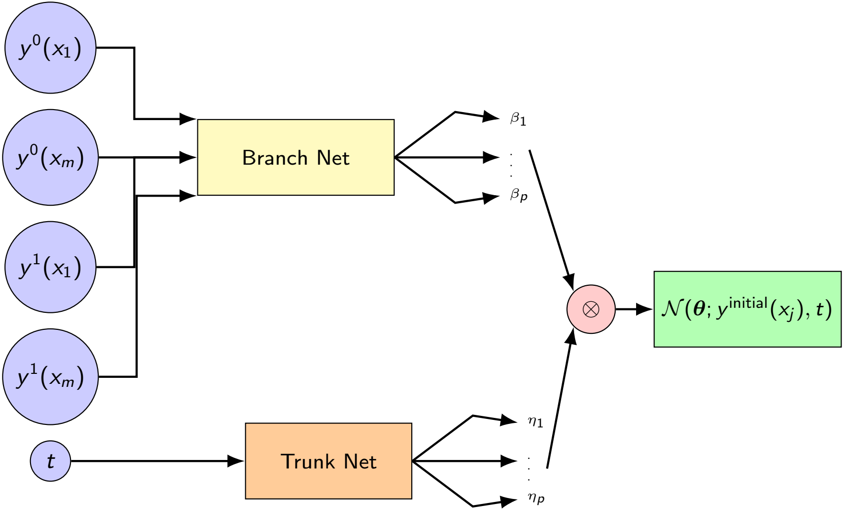
```
    X_train = (X_train[:, :-1], X_train[:, -1:])
    X_test = (X_test[:, :-1], X_test[:, -1:])

    data = dde.data.Triple(
    X_train=X_train, y_train=y_train, X_test=X_test, y_test=
        y_test
    )
```

Lu, L., Jin, P., Pang, G., Zhang, Z., Karniadakis, G.E.:*Learning nonlinear operators via deeponet based on the universal approximation theorem of operators.* **Nature Machine Intelligence 3**(3), 218-229, 2021.

DeepONet. Experiment 4: fitting the deeponet model

Next, we select the architecture of the DeepONet



DeepONet. Experiment 4: fitting the deeponet model

The implementation in DeepXDE is as follows:

```python
m = X_train_input_dimension  # inferred from training data

dim_t = 1  # input dimension of the trunk

p = 10  # output dimension of the branch and trunk nets

net = dde.nn.DeepONet(
[m, 40, p],        # dimensions of the branch net
[dim_t, 40, p],    # dimensions of the trunk net
"relu",            # activation function
"Glorot normal",   # initialization of parameters
)

model = dde.Model(data, net)

model.compile("adam", lr=0.001)

losshistory, train_state = model.train(iterations=ITERATIONS)

dde.saveplot(losshistory, train_state, issave=True, isplot=True)
```

DeepONet. Experiment 4: Smooth initial conditions

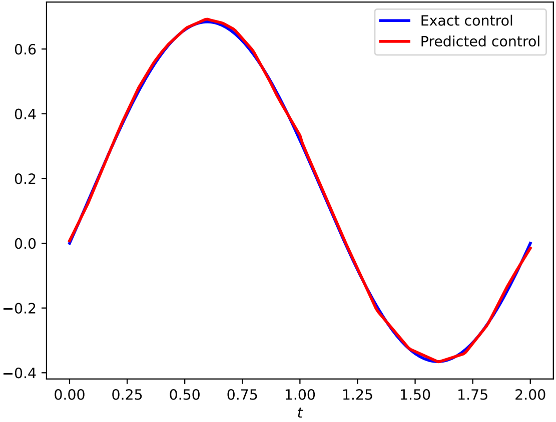


Figure: Experiment 4: wave equation. Exact versus predicted solutions for the smooth initial conditions $y^0(x) = y^1(x) = \sin(\pi x)$. $n_{functions} = 10^4$, $(\ell_0, \ell_1) = (0.25, 0.125)$, $n_{sensors} = 101$, $p = 40$. Relative error $\approx 1\%$.

DeepONet. Experiment 4: Unsmooth initial conditions

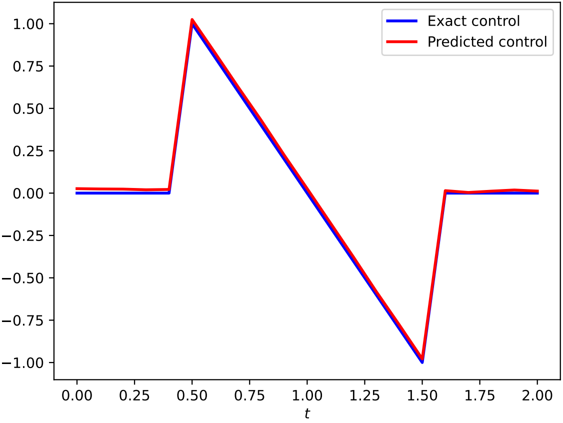


Figure: Experiment 4: wave equation. Exact versus predicted solutions for the unsmooth initial conditions $y^0(x) = \begin{cases} 4x, & 0 \leq x \leq 0.5 \\ 0, & 0.5 < x \leq 1 \end{cases}, \quad y^1(x) = 0.$

$n_{functions} = 10^4$, $(\ell_0, \ell_1) = (0.03125, 0.03125)$, $p = 100$. $n_{sensors} = 11$. Relative error $\approx 4\%$.

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Thank you for your attention !